MCBB3.0 - AWS

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Version Control

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KB Info

MCBB 3.0 is Fiserv's next-generation cloud networking product. Designed to work seamlessly with the new Global Backbone, MCBB 3.0 is set to replace all existing cloud networking solutions currently in use at Fiserv.

The product is engineered to be scalable, easy to deploy and operate, and rich in features. To formalize its capabilities, a baseline requirements list has been compiled and made available for vendor review.

While it's understood that not every design requirement may be fully met, vendors are strongly encouraged to propose alternative solutions to address any gaps. These proposals may include leveraging third-party products owned or certified by Fiserv, as well as utilizing internal Fiserv capabilities such as automation.

The scope of the solution as a minimum has to cover the following CSP regions:

- eu-west-1 (Ireland)
- eu-central-1 (Frankfurt)
- ap-southeast-1 (Singapore)
- ap-southeast-2 (Sydney)
- ap-south-1 (Mumbai)
- ap-south-2 (Hyderabad)
- us-west-2 (Oregon)
- us-east-1 (N. Virginia)
- us-east-2 (Ohio)
- me-central-1 (UAE)
- sa-east-1 (Sao Paulo)

The geo regions not covered in the list, but desired to be in scope of the solution:

- Argentina
- Panama
- Mexico

Fiserv owns multiple Cloud OnRamp circuits across various regions. The solution must support short-path routing from each region to the nearest domestic on-premises WAN or, if unavailable, to the closest regional on-ramp. The Cloud OnRamp WAN may consist of dedicated circuits ranging from 10 to 100 Gbps, or hosted WAN capacity provisioned through existing or new carriers. Traffic within AWS should be routed internally within the cloud service provider (CSP).

All network traffic must be encrypted. That requirement includes either CSP backbone or Cloud OnRamp of any scale

The solution must either provide, or clearly indicate the availability of, low-latency connectivity options with other CSPs within the same region.

The Fiserv Cloud network comprises multiple routing domains with overlapping address spaces. The proposed MCBB3.0 solution must support 1:1 network address translation to enable communication between overlapping segments.

Route propagation should be granular down to specific prefixes, with configurable options to allow, deny, or automatically aggregate routes to broader subnets where applicable. All routes must be tagged with Fiserv-required BGP communities. Routing tables across the CSP network must be fully visible and accessible via API, CLI, or UI. All routing policies must be explicitly controlled by Fiserv, and the CSP must not inject unsolicited routes into the routing tables.

The proposed solution must offer a consistent feature set across all regions. Any limitations in features or scalability must be clearly defined and aligned with Fiserv-specific requirements.

The solution must not include hidden components, such as inline load balancers, that could potentially impact Fiserv traffic.It is highly desirable if the new solution offers a level of bandwidth protections against applications with bulk consumption.

Network telemetry is expected to provide Fiserv with real-time visibility into traffic flows, top talkers, fat flows, and related metrics. Additionally, network-related data should be retrievable from the CSP infrastructure to support internal capacity planning and alerting.

Figure 1. MCBB3.0 High level diagram

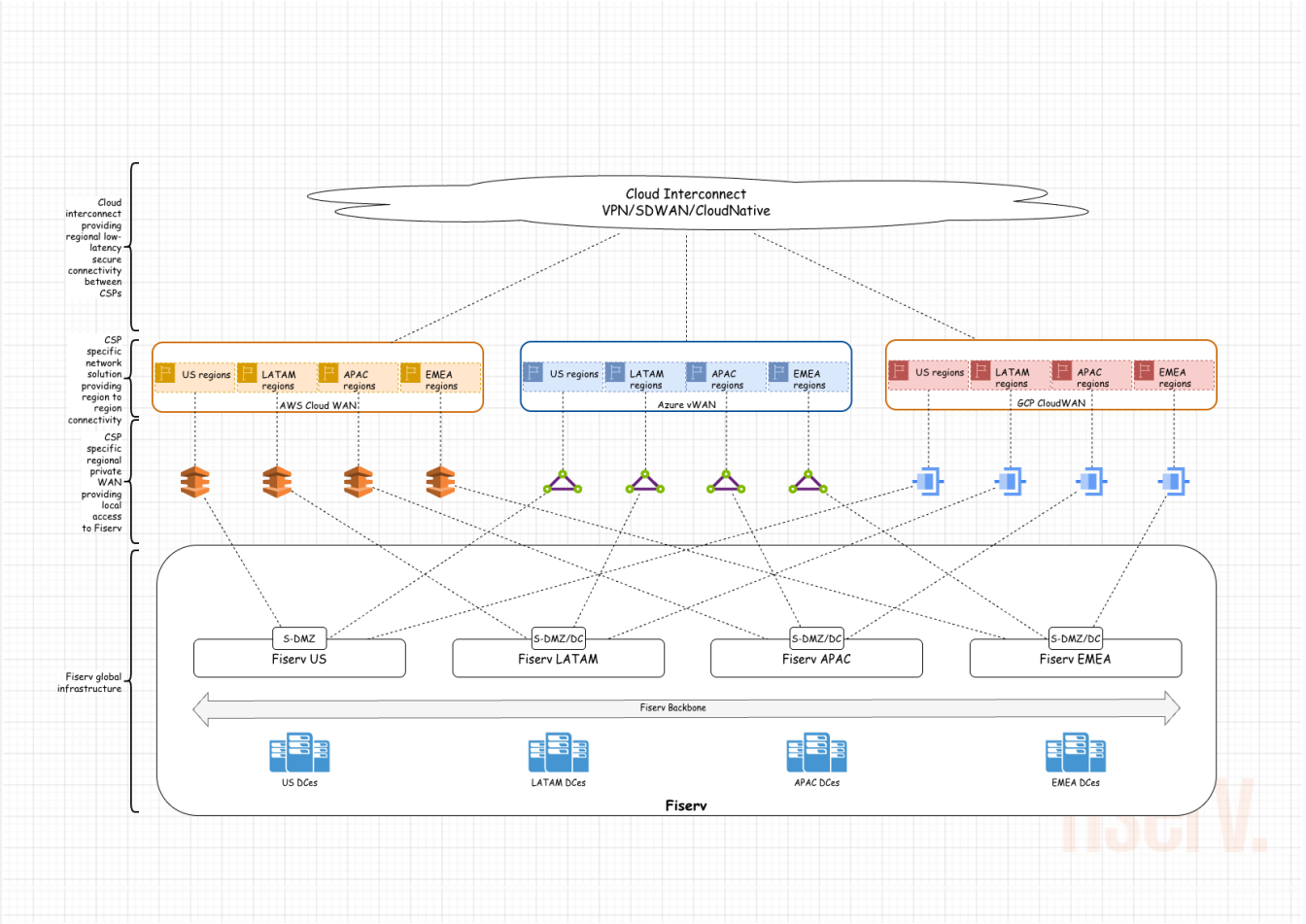


Figure 2. MCBB3.0 refined baseline design requirements

DR N	Description	Comments
	Transport	
DR 1.1	Connectivity between VPC/VNETs of the same Cloud Service Provider (CSP) within the region.	
DR 1.2	Connectivity between VPC/VNETs across different regions of the same CSP.	
DR 1.3	Connectivity between VPC/VNETs to on-premises systems using the nearest Fiserv Cloud entry point in the same geographic region (geo affinity)	

DR 1.4	Connectivity between VPC/VNETs to on-premises systems using the closest Fiserv Cloud entry point if the local entry point is not available.	
DR 1.5	Connectivity between CSPs and other CSPs using native cloud networking and/or other solutions via private or public network infrastructure.	
DR 1.6	Enforcement of network security policies to control network access per VPC/VNET region, compliant with Fiserv Cyber and governance requirements.	
DR 1.7	Cloud OnRamp scalability ranging from sub-gig to 100g	
DR 1.8	Inter Cloud connectivity scalability ranging from sub-gig to 100g	
DR 1.9	Support for multi-pathing (Equal-Cost Multi-Path Routing, ECMP).	
DR 1.10	~~Network encryption support across Cloud OnRamp of any scale aligns with Fiserv Cyber requirements.~~	Cyber decision on network encryption
DR 1.11	~~Network encryption across CSP boundaries (with evidence provided at any time).~~	Cyber decision on network encryption
DR 1.12	~~Cross regional data forwarding with the avoidance of specific geopolitical boundaries~~	self-imposed requirement
Routing		
DR 2.1	Route propagation control at the regional level.	
DR 2.2	Route propagation control at the cross-regional level.	

DR 2.3	Route propagation control at the Cloud OnRamp level.	
DR 2.4	Route propagation control for inter-cloud connectivity.	
DR 2.5	Granularity of routing controls per prefix utilizing BGP communities.	
DR 2.6	Route table scalability supporting up to Xk routes across all the elements of the solution	X is defined by the default limits imposed by each CSP
DR 2.7	Sub-second convergence time in case of blackout conditions.	
DR 2.8	Network convergence during brownout conditions (e.g., when packet latency exceeds the threshold or sporadic packet drops occur).	self-imposed requirement
DR 2.9	Granular end-to-end visibility on routing tables.	
DR 2.10	Multi-tenancy capabilities similar to Virtual Routing and Forwarding (VRF).	
Observability		
DR 3.1	Real-time telemetry on active traffic flows using standard tools/API calls.	self-imposed requirement
DR 3.2	Real-time reporting of SLA metrics (latency/packet loss).	self-imposed requirement to be addressed through third-party services like 1ke
DR 3.3	Historical telemetry on network flows, including top talkers, top requestors, and primary flows.	
DR 3.4	Performance metrics reporting utilising push, pull, or subscription mechanisms.	
Miscellaneous		

DR 4.1	Real-time provisioning using Infrastructure as Code (IAC) such as Terraform, CI/CD, etc.	
DR 4.2	~~Granular Network Address Translation (NAT) capabilities (1:1, N:1)~~	self-imposed requirement to be reviewed and likely withdrawn in favour of supporting a single GRT with following migration of existing subs /account
DR 4.3	~~Granular Network Address Translation (NAT) capabilities (1:1, N:1) between tenants within the same CSP~~	self-imposed requirement to be reviewed and likely withdrawn in favour of supporting a single GRT with following migration of existing subs /account
DR 4.5	Quality of Service (QoS) for guaranteeing bandwidth capacity at the Cloud OnRamp level.	Not supported by CSP in egress (C to F)
DR 4.6	Policy-based routing for specific applications or flows.	self-imposed requirement
DR 4.7	Packet sniffing capabilities accessible via UI/API.	self-imposed requirement
DR 4.8	Integration on the routing level with NVAs such as non cloud native firewalls, S2S gateways, SDWAN appliances	
DR 4.9	Flexible service chaining capability to divert traffic to the network security /etc appliances	
DR 4.10	Cloud OnRamp integrated to S-DMZ design	

Reading list

The reading list on C-WAN (from the Fiserv-AWS techical call):

- CloudWAN and DX integration: https://aws.amazon.com/blogs/networking-and-content-delivery/simplify-global-hybrid-connectivity-with-aws-cloud-wan-and-aws-direct-connect-integration/
- CloudWAN availability in regions: https://docs.aws.amazon.com/network-manager/latest/cloudwan/what-is-cloudwan.html
- CloudWAN Quotas: https://docs.aws.amazon.com/network-manager/latest/cloudwan/cloudwan-quotas.html#cloudwan-routing
- DirectConnect Quotas: https://docs.aws.amazon.com/directconnect/latest/UserGuide/limits.html
- CloudWAN CloudWatch Metrics : https://docs.aws.amazon.com/network-manager/latest/cloudwan/cloudwan-metrics.html
- Migration Guide : https://aws.amazon.com/blogs/networking-and-content-delivery/aws-cloud-wan-and-aws-transit-gateway-migration-and-interoperability-patterns
- ReInvent C-WAN slidedeck: https://reinvent.awsevents.com/content/dam/reinvent/2024/slides/net/NET311_Build-scalable-secure-global-connectivity-with-AWS.pdf

Cloud WAN based design

Overview

AWS Cloud WAN is preferred solution to establish connectivity to/between/from Fiserv sites and AWS regions. There is a number of use cases to be supported, explored, tested, and documented in the MCBB 3.0 design.

The following baseline assumptions must be accounted for:

- The US environment is large-scale, utilizing 10–100GE on either BB or Cloud OnRamp.
- The US environment supports colocation with NaaS providers such as Megaport or Equinix.
- The US Cloud OnRamp is preferably based on dedicated Direct Connect links.
- The US BB is a multisite network connecting data centers and edge locations.
- The US Cloud OnRamp is preferably deployed at edge locations.
- The international environment is small to medium scale, typically using sub-GE, 1GE, or 10GE.
- A standard international deployment consists of two data centers within the same country, either in metro or national proximity.
- International environments generally do not have a BB; instead, they rely on DCI links of sub-10GE or 10GE+.
- International sites are typically not colocated with AWS Edge sites in the same data center facility.
- Third-party WAN/NaaS (if available) must be used for most international sites to establish Direct Connect links with AWS.

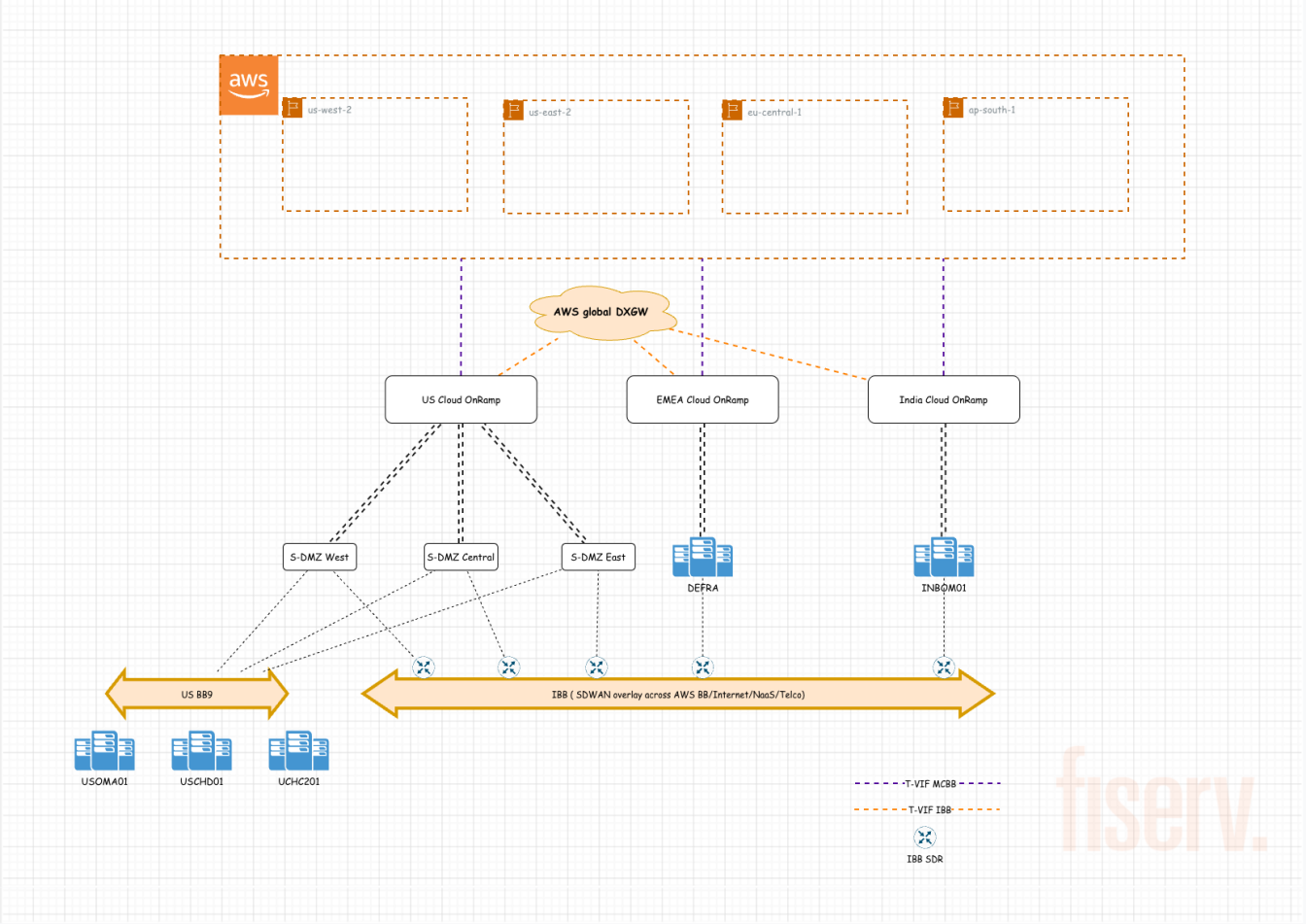
- Third-party WAN may be provided as L2 or L3 VPN by domestic telcos.
- In most cases, international Direct Connect links are hosted connections with sub-1GE capacity.

Cloud WAN with Split DXGW Design

The Cloud WAN (C-WAN) segment is extended across all AWS regions, with local VPCs attached to Cloud Network Edges (CNEs). Each region includes a set of domestic Direct Connect (DX) connections delivered to various DX edge sites.

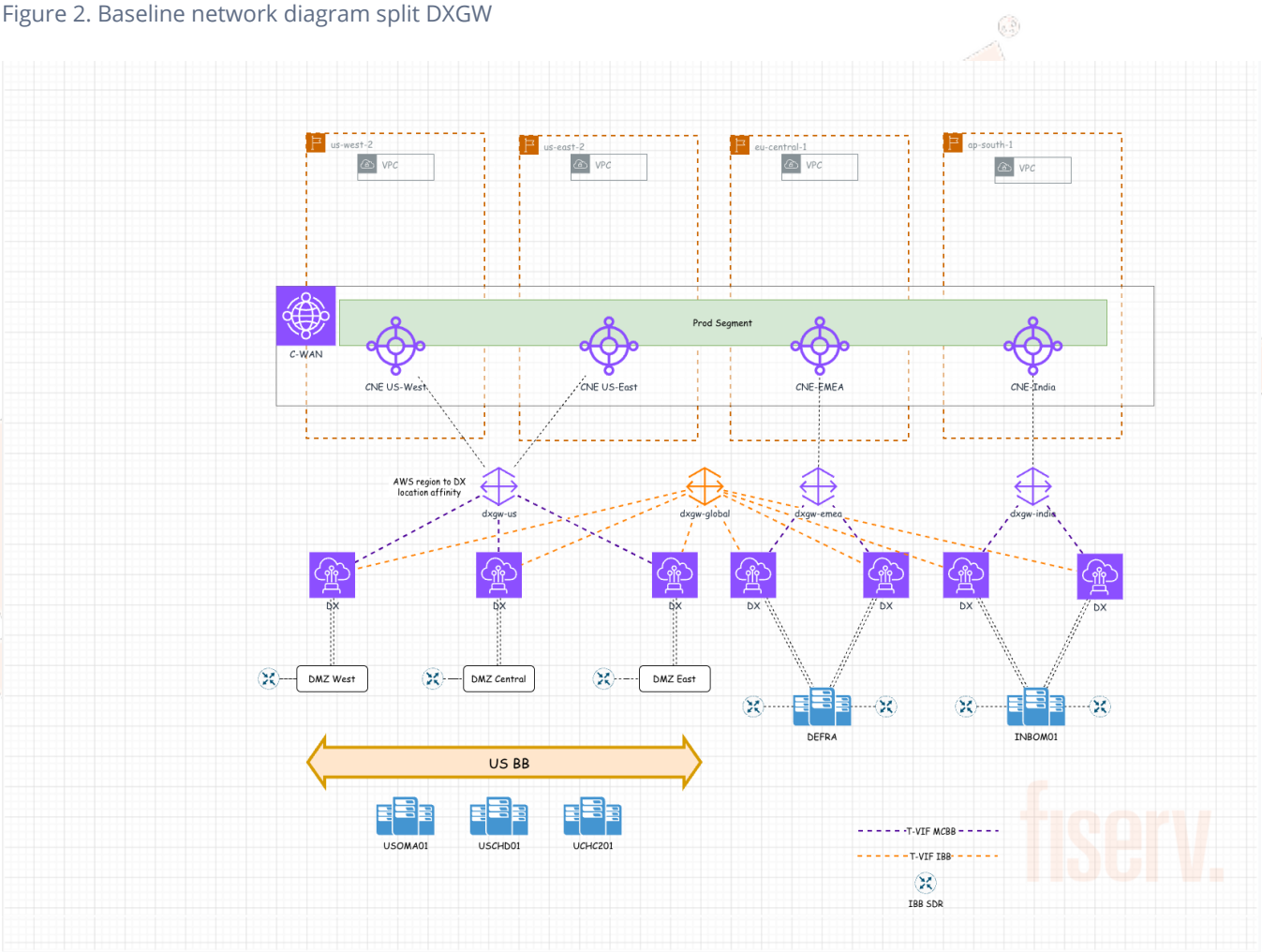
AWS supports affinity between DX locations and AWS regions, as detailed at https://aws.amazon.com/directconnect/locations/. By default, an AWS region learns on-premises routes from AWS DX through its associated regional DX location.

Figure 2. High level MCBB/IBB design



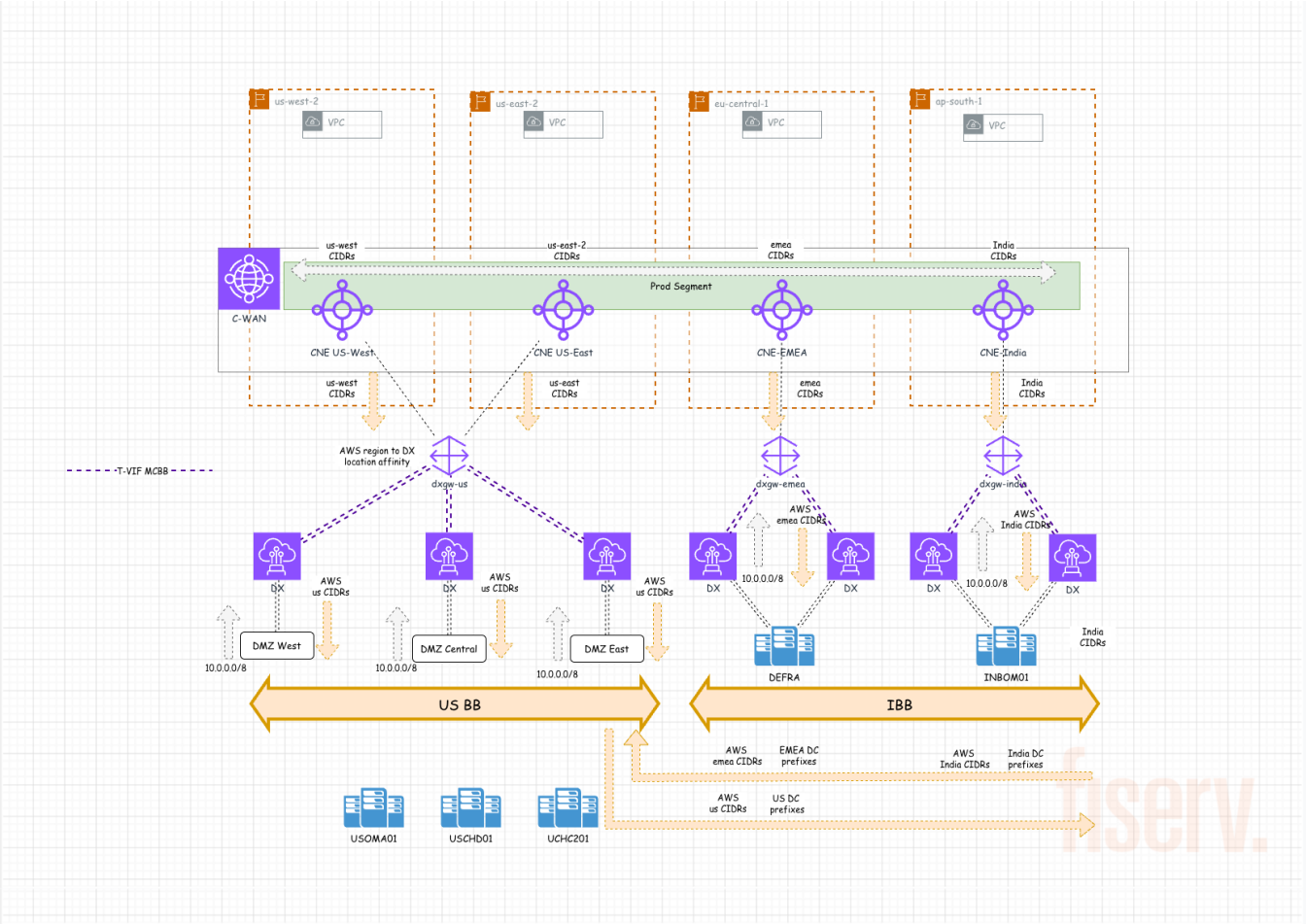
If a regional DX location is unavailable or fails, AWS Direct Connect Gateway (DXGW) can redirect traffic to other DX connections attached to the same DXGW. When multiple alternative DX connections are available, traffic is distributed across them using Equal-Cost Multi-Path (ECMP) routing. This behavior necessitates a design decision to limit the scope of DXGW to regional boundaries. Doing so prevents scenarios where AWS regions might attempt to route traffic through DX connections located in different geopolitical areas. AWS solution addressing these scenarios is to use dedciated DXGW per region

Figure 2. Baseline network diagram split DXGW



The C-WAN segment does not propagate local VPC routes to remote DXGW attachments. Instead, route propagation is limited to CNE-to-CNE communication and to the DXGW associated with the local CNE. Consequently, the DXGW serves both as a route aggregator and as the data plane between different C-WAN regions when forwarding traffic to on-premises locations. From a technical perspective, all routes ingested by CNEs into the DXGW are advertised to every DX connection attached to that DXGW.

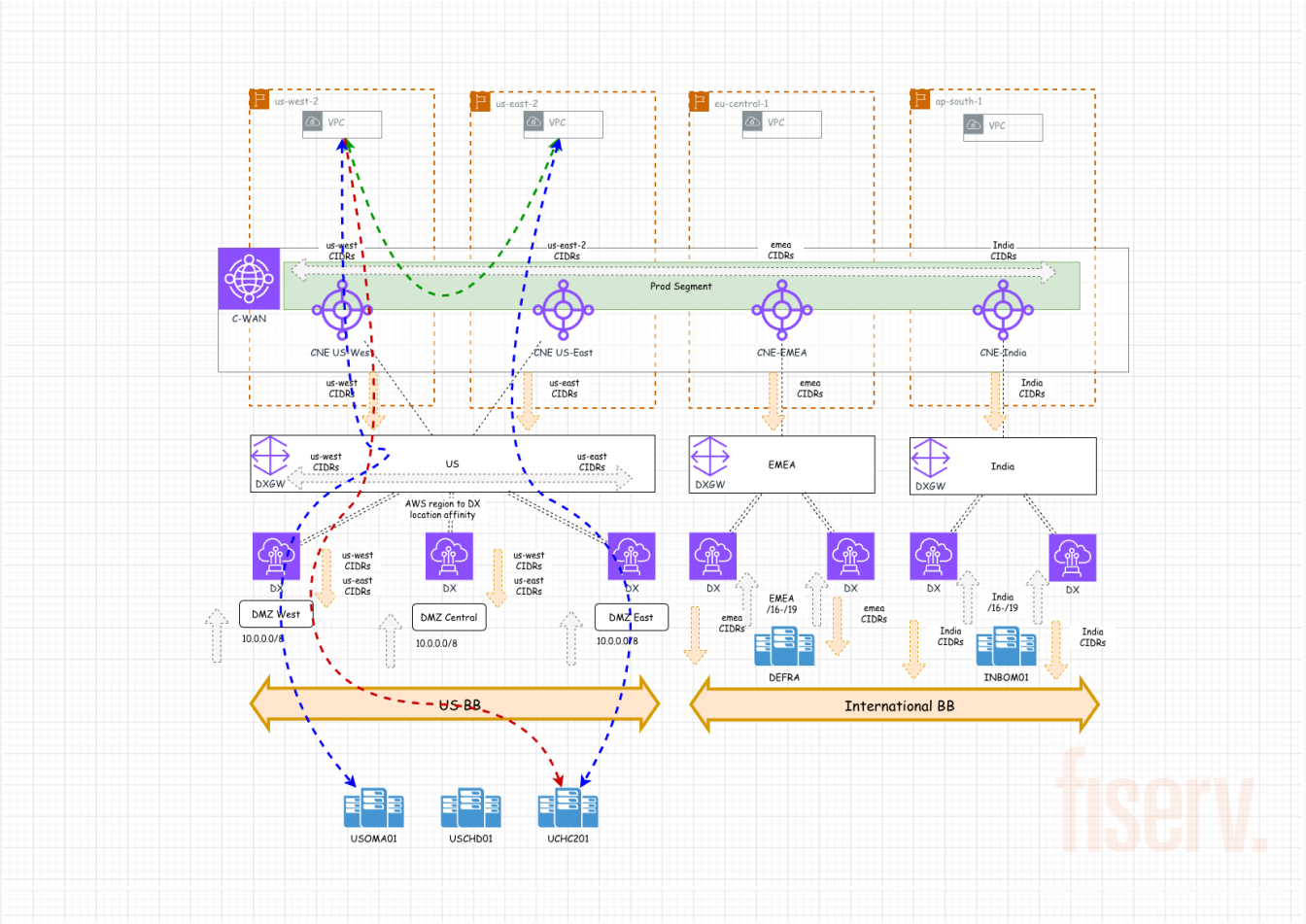
Figure 3. Baseline network diagram split DXGW route propagation



All the traffic flows of VPCes connected to the same C-WAN segment will be able to communicate to each other using C-WAN dataplane acros AWS BB.

Traffic of US regions will be also able to leverage DXGW to have optimal path towards the closest Fiserv Cloud OnRamps locations with the fall back options to other DXes in US.

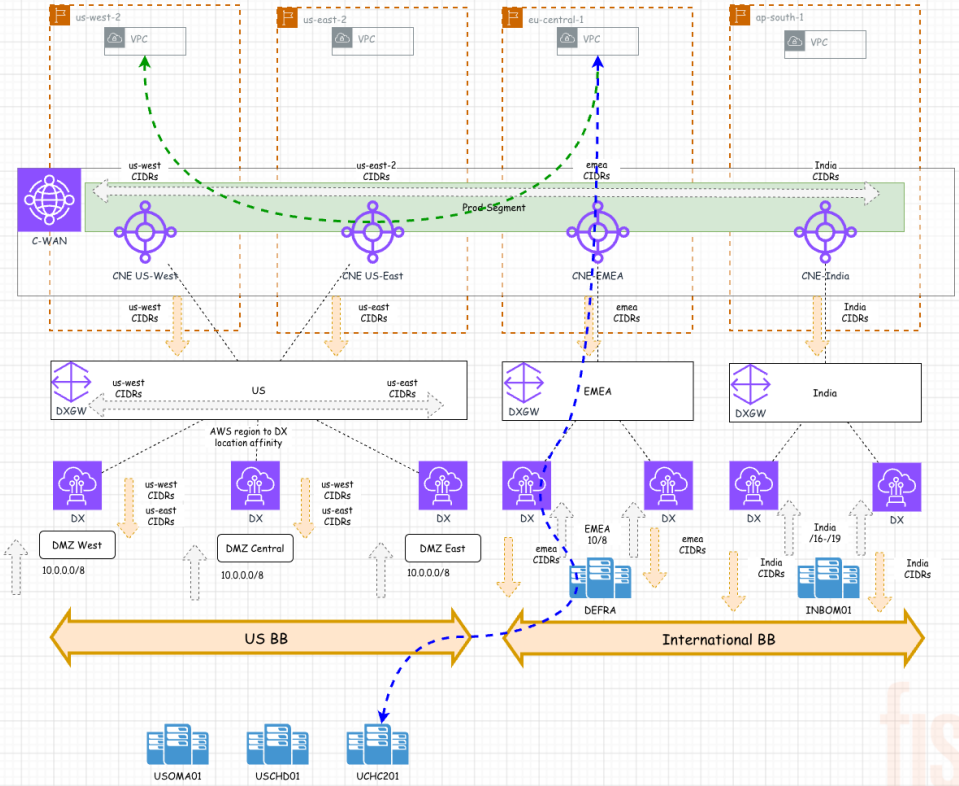
Figure 4. Baseline network diagram split DXGW US traffic flows



Traffic crossing the international boundaries have to be processed by Fiserv International Backbone (IBB)

IBB is planned as an SDWAN overlay connecting Fiserv OnPrem sites over the secure tunnels established across CSP backbones. In case of AWS that techically means a separate global DXGW or a Public VIF preferably established across the same Cloud OnRamps used by MCBB

Figure 4. Baseline network diagram split DXGW INT traffic flows


us-west-2
us-east-2
eu-central-1
ap-south-1
VPC
us-west CIDRs
us-east-2 CIDRs
emea CIDRs
India CIDRs
Prod-Segment
C-WAN
CNE US-West
CNE US-East
CNE-EMEA
CNE-India
us-west CIDRs
us-east CIDRs
emea CIDRs
India CIDRs
DXGW
US
EMEA
India
AWS region to DX location affinity
DX
DMZ West
DMZ Central
DMZ East
10.0.0.0/8
EMEA 10/8
India /16-/19
DEFRA
INBOM01
US BB
International BB
USOMA01
USCHI01
UCHC201